

Timothy Lo

Researcher · Developer · Medical Imaging · Artificial Intelligence

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Education

Ph.D. in Medical Physics and Biomedical Engineering 2024 - Present

University College London

In Progress

Thesis: Developing a software suite to analyse the potential benefits of upright radiotherapy.

M.Res. in Artificial Intelligence and Machine Learning 2022 - 2023

Imperial College London

Distinction

Research Project: Automatic segmentation and diagnosis for ovarian masses.

B.Sc. in Computer Science 2019 - 2022

King's College London

First Class Honours

Publications

Technical Note

A low-cost, flexible, and accurate technique to study new immobilisation systems for radiotherapy and their 2D in-plane positional reproducibility and stability

Physics in Medicine and Biology, Vol. 71

2026 · DOI: 10.1088/1361-6560/ae6afa

Journal Article

Upright radiotherapy for breast cancer: a pre-clinical study considering photon and proton beam access, plus arm positioning

Frontiers in Oncology, Vol. 15

2025 · DOI: 10.3389/fonc.2025.1668109

Conference Presentations

ESTRO 2026 (Oral Presentation - Proffered Paper, 2026)

SPLASH: Semi-automatic Pipeline for Lung motion Analysis using Single-frame Human input

ESTRO 2026 (Poster Presentation, 2026)

Anatomical Analyses for the Prostate Region in Upright and Supine Postures

Upright Consortium 2026 (Oral Presentation, 2026)

GUAVA: a Geometric Upright-supine Anatomical Variation Analysis framework

AAPM 2025 (Poster Presentation, 2025)

A Deformable Registration-based Pipeline for the Characterisation of Respiratory Motion from 2D Lung Cine MRI

ESTRO 2025 (Oral Presentation - Poster Discussion, 2025)

A Low-cost, Flexible, and Accurate Technique to Study New Immobilisation Systems for Radiotherapy and their Associated Positional Reproducibility/Stability

Upright Consortium 2024 (Oral Presentation - Proffered Mini Oral Presentation, 2024)

Computer Vision-based Marker Detection for Quantitative Assessment of Upright Position Reproducibility

Professional Experience

Post Graduate Teaching Assistant

Sep 2024 - Current

University College London, United Kingdom

- Preparing and delivering course material.
 - Exam and coursework marking.
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Coding Instructor

Feb 2024 - Feb 2025

Code Kids Robotics Ltd., United Kingdom

- Hosted weekly online teaching sessions.
 - Taught in 5-day camps with intensive teaching blocks.
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Programming Tutor

Sep 2023 - Current

Online - United Kingdom

- Hosted weekly online teaching sessions.
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Software Engineer

Apr 2022 - Jul 2022

Pawsible, Hong Kong

- Delivered a proof of concept application that communicates with a sensory device that takes in vital readings.
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Robotics Club Chairman

Sep 2016 - Sep 2018

HKCCCU Logos Academy, Hong Kong

- Lead and took part in a team to win two World Championships in Robofest GRAF 2017.
 - Facilitated more teams to win 1st and 2nd runner ups in Robofest 2017 and 2018.
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Technical Skills

Programming: Python, JavaScript / TypeScript, C / C++, SQL

Libraries: NumPy / Pandas, PyTorch, Scikit-learn, TensorFlow, Matplotlib / Seaborn, Manim

Tools: Jupyter, React.JS, Next.JS, Node.JS, REST API, Firebase, Git / GitHub, Linux / Bash, LaTeX, Adobe Creative Suite, Google Cloud Computing

Domains: Machine Learning, Deep Learning, Computer Vision, Image Processing, NLP, Statistical Analysis, Data Visualisation, Software Engineering, Mobile Development, Web Development, Research Methods, Scientific Writing